MILITARY SPECIFICATION

CHARGER, BATTERY, PP-7806/BRT-6

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance, manufacture, and test requirements for the Charger, Battery, PP-7806/BRT-6, hereinafter referred to as the BC, and the Interconnecting Cable Assembly, CX-13318/BRT-6, hereinafter referred to as the Cable Kit. The BC and the Cable Kit are part of the Buoy Transmitting Set, AN/BRT-6.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbook form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR 003-121), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5820

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SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment And Systems, Requirements For
MIL-P-13949	Plastic Sheet, Laminated, Metal Clad (For Printed Wiring Boards), General Specification For
MIL-E-16400G	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Provisioned Items (Repair Parts): Packaging Of
MIL-F-18264	Finishes: Organic, Weapons System, Application And Control Of
MIL-S-19500	Semiconductor Devices, General Specification For
MIL-C-21097	Connectors, Electrical, Printed Wiring Board, General Purpose, General Specification For
MIL-C-24643/45	Cable, Electrical, 300 Volts, Type LS2U (Including Variation LS2UA)
MIL-C-28840	Connectors, Electrical, Circular Threaded, High Shock, High Density, Shipboard, Class D, General Specification For
MIL-B-29513	Buoy, Radio Transmitter, T-1455/BRT-6
MIL-M-38510	Microcircuits, General Specification For
MIL-P-55110	Printed Wiring Boards, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)

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MIL-STD-275	Printed Wiring For Electronic Equipment
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/ Demonstration/Evaluation
MIL-STD-781	Reliability Testing For Engineering Development, Qualification, And Production
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-965	Parts Control Program
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities
MIL-STD-1683	Connectors And Jacketed Cable, Electric, Selection Standard For Shipboard Use
DoD-STD-1686	Electrostatic Discharge Control Program For Protection Of Electrical And Electronic Parts, Assemblies And Equipment (Excluding Electrically Initiated Explosive Devices) (Metric)
MIL-STD-2164	Environmental Stress Screening Process For Electronic Equipment
MS 27505	Connector, Receptacle, Electrical Back Panel, Box Mounting Flange, Crimp Type, Bayonet Coupling, Series 1

HANDBOOK

MILITARY

DoD-HDBK-263	Electrostatic Discharge Control Handbook For Protection Of Electrical And Electronic Parts, Assemblies And Equipment (Excluding Electrically Initiated Explosive Devices) (Metric)

2.1.2 Government drawings and publications. The following Government drawing and publication form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

F53711-409-6379608	Remote Indicator And Interconnecting Cables

PUBLICATION

NAVAL AIR SYSTEMS COMMAND (NAVAIR)

AS-4613	Application And Derating Requirements For Electronic Components, General Specification For

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The BC and Cable Kit shall be in accordance with MIL-E-16400G, to the extent specified herein.

3.1.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.2 BC description. The BC is part of the Buoy Transmitting Set, AN/BRT-6 (see FIGURE 1). The BC provides a charging station for up to six Radio Transmitter Buoys, T-1455/BRT-6 (buoys) specified in MIL-B-29513. The BC provides a high and low charging voltage to the buoys, displays the accummulated power-on elapsed time and monitors the buoy charge current and voltage.

3.1.3 Cable Kit description. The Cable Kit consists of one cable assembly which provides the electrical interconnection between the BC and six buoys.

3.2 Performance. BC performance shall be as specified in 3.2.1 through 3.2.5.

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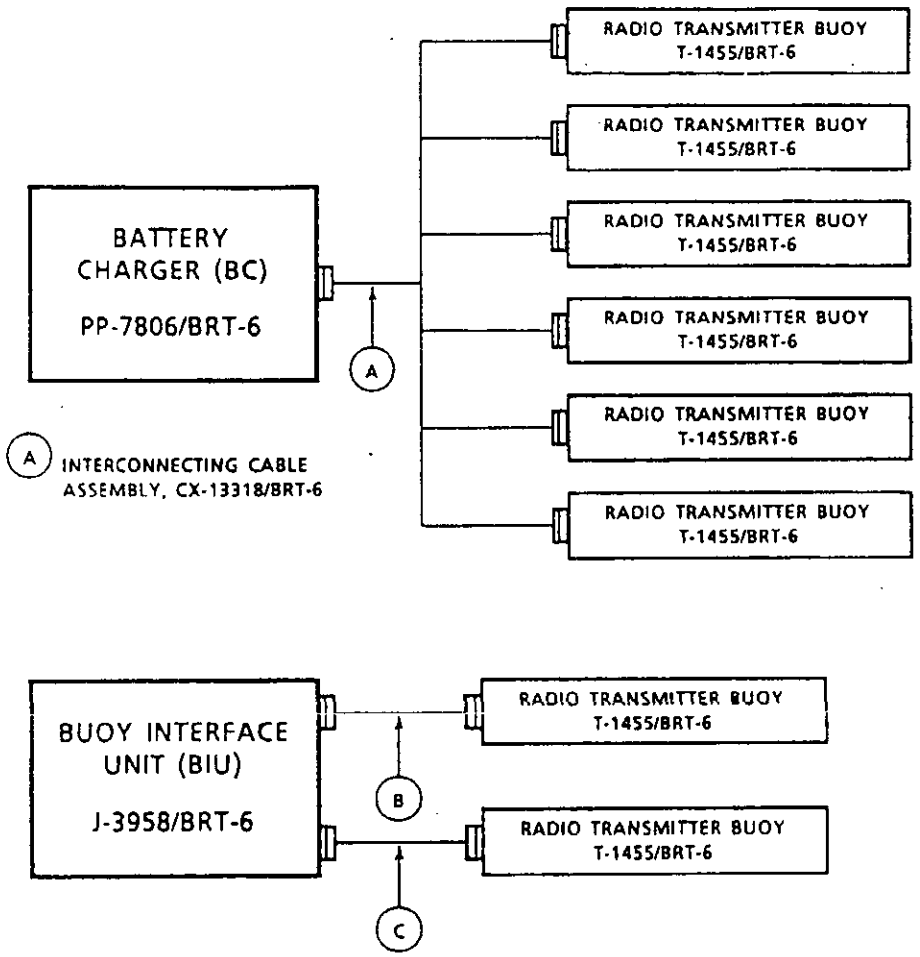


FIGURE 1. Buoy Transmitting Set, AN/BRT-6 block diagram.

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3.2.1 Prime power. Prime power shall be 115 volts alternating current (VAC), 60 hertz (Hz), single-phase, Type I, in accordance with DoD-STD-1399, Section 300. The power is connected via J1 as shown in FIGURE 2. Total current draw shall not exceed 2 amperes (amp). The alternating current (AC) power switch shall be a two-position switch located on the front panel and shall be labeled with ON and OFF designations. A power indicator shall be located on the front panel and shall be labeled POWER ON. The power indicator shall be connected across the input power conductors on the load side of the switch. The AC input fuses, F1 and F2, shall be located on the front panel and labeled with their corresponding values. AC input lines shall be fused at the load side of the power switch as shown in FIGURE 2 and shall have blown fuse indicators. Spare fuses shall be located within the BC.

3.2.2 Elapsed time meter. An elapsed time meter, M2, shall be located on the front panel and labeled OPERATING. M2 shall display accumulative on-time of prime power. The M2 fuse, F3 (see FIGURE 2), shall be located on the front panel, labeled with its value and have a blown fuse indicator.

3.2.3. Charger outputs. The BC shall provide 12 charging outputs consisting of one high (HI) and one low (LO) voltage output for each of the six buoys in accordance with FIGURE 2. The HI voltage charging output shall be 65 volts direct current (VDC) ±10 volts (V) open-circuit voltage and shall be a constant current source of 50 milliamperes (mA) ±5 mA. The LO voltage charging output shall be 30 VDC ±7 V open-circuit voltage and shall be a constant current source of 150 mA ±15 mA.

3.2.4 Charging meter. Meter M1 shall be located on the front panel and shall be used to measure voltage and current of each of the 12 charger outputs (see 3.2.3). The meter scale shall have linear black divisions with full scale values of 35 V, 75 V, 75 mA, and 225 mA. The scale background coloring shall be red from 0 percent to 50 percent of full scale, green from 50 percent to 75 percent of full scale, and white from 75 percent to 100 percent of full scale. The remainder of the meter scale face shall be white. The overall accuracy of the meter shall be 5 percent.

3.2.4.1 Buoy select switch. A 12-position switch shall be located on the front panel and labeled BUOY SELECT with a position designation for each charger output. The BUOY SELECT switch shall be used to select which one of the 12 charging outputs will be measured by meter M1.

3.2.4.2 Meter select switch. A two-position switch shall be located on the front panel and labeled METER SELECT with two positions labeled CURRENT and VOLTAGE. The METER SELECT switch shall be used to select whether voltage or current is measured by meter M1.

3.2.5 Leakage current. The leakage current shall not exceed 5 mA.

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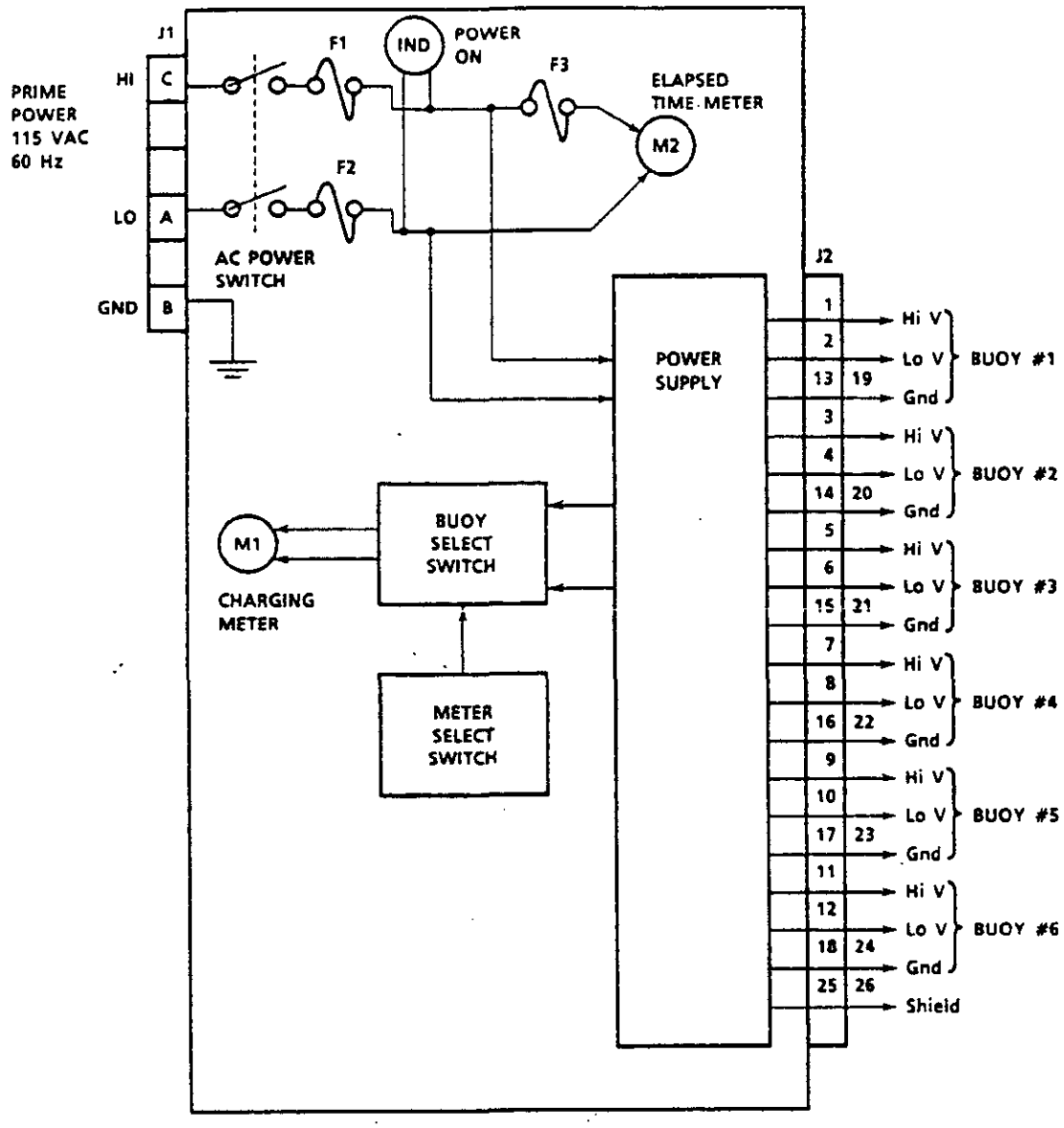


FIGURE 2. BC.

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3.3 Physical characteristics. The physical characteristics shall be as specified in 3.3.1 through 3.3.7.

3.3.1 Dimensions and weight. The BC shall consist of a front panel, enclosure cabinet, and mounting plate. The BC shall conform to the dimensions shown in FIGURE 3. The total weight of the BC shall not exceed 15 pounds (lbs). Total weight of the Cable Kit shall not exceed 30 lbs and the length shall not exceed 30 feet.

3.3.2 Front panel. The BC front panel shall be hinged to the enclosure and, in the closed position, shall become the front surface of the enclosure. The front panel shall attach to the enclosure by captive screws. Front panel controls, indicators, and labels shall be in accordance with FIGURE 4.

3.3.3 Mounting plate heat sink. The mounting plate shall be the rear surface of the enclosure and shall be used as a heat sink so that forced air cooling is not necessary.

3.3.4 Human engineering. The BC shall conform to the human engineering operational and maintenance design criteria of MIL-STD-1472. The design of the operator and maintainer panel layouts and controls shall conform to 3.3.2 and MIL-STD-1472.

3.3.5 Accessibility. Accessibility shall be in accordance with MIL-STD-454, Requirement 36. An automatically operated lock shall be provided to lock the access door in the servicing position.

3.3.6 Cables and connectors. Cables and connectors shall be as specified in 3.3.6.1 through 3.3.6.4.

3.3.6.1 Cable Kit. The Cable Kit shall be in accordance with NAVSEA Drawing F53711-409-6379608 and MIL-C-24643/45.

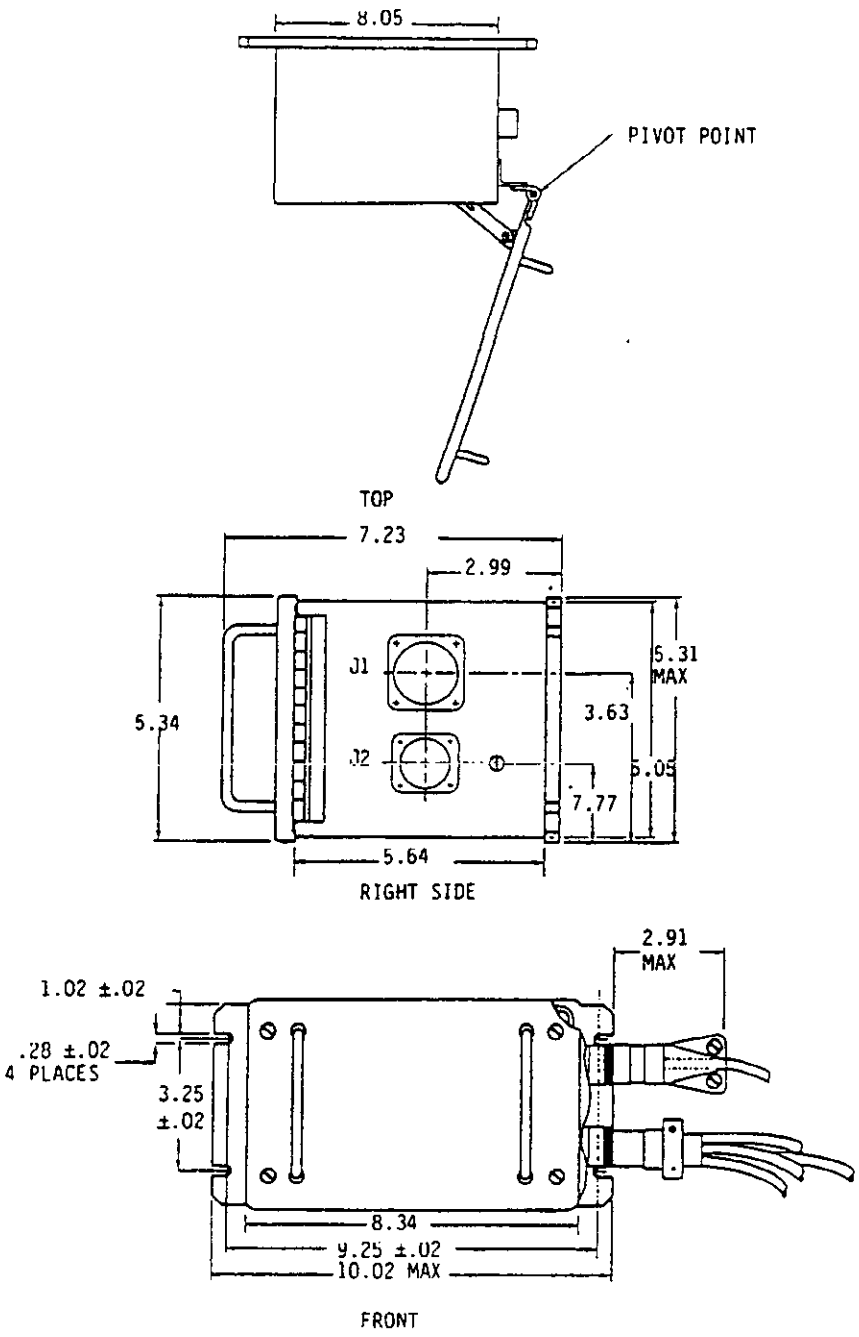
3.3.6.2 BC internal cabling. Sufficient cable shall be provided internal to the enclosure to accommodate front panel opening. The cables shall be arranged and guided to prevent snagging or interference during front panel extension.

3.3.6.3 Connectors. All external electrical connectors shall be located on the right side of the enclosure and shall be in accordance with MS 27505, the types specified in TABLE I, MIL-STD-1683, and MIL-C-28840.

TABLE I. External connectors.

Connector	Type
J1	MS 27505E-17B-6P
J2	MS 27505E-15B-35SA

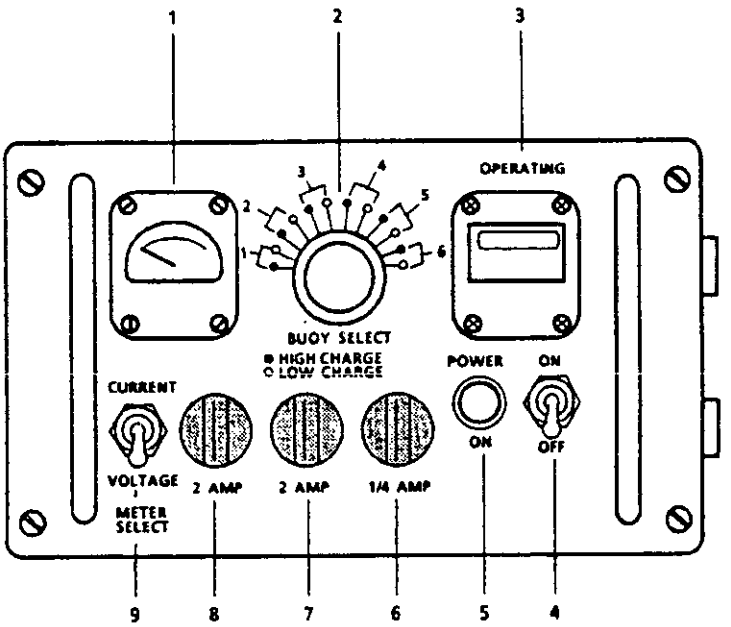
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Note: 1. Unless otherwise specified, all tolerances are ±0.05.
2. All dimensions are in inches.

FIGURE 3. BC dimensions.

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1. VOLT/AMP METER (M1)
2. BUOY SELECT SWITCH
3. OPERATING METER (M2)
4. POWER ON/OFF
5. AC POWER ON INDICATOR
6. 1/4 AMP FUSE (F3)
7. 2 AMP FUSE (F2)
8. 2 AMP FUSE (F1)
9. METER SELECT SWITCH

FIGURE 4. BC controls and indicators (front view).

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3.3.6.4 Power connections. 115 VAC power cable connections within the BC shall have the color and conductor assignments specified in TABLE II. Pin designations shall be as specified in TABLE II. Conductor insulation color shall be maintained from the BC input connections to all components with the same voltage and frequency as the input power.

TABLE II. 115 VAC equipment power connections.

Pin designation	Conductor assignment	Insulation color
A	115 VAC return	White
B	Safety ground	Green
C	115 VAC HI (supply)	Black

3.3.7 Power interlock. A bypassable interlock shall be provided to disconnect all power to the BC when the front panel is open. When this interlock is bypassed, an internal lamp shall illuminate.

3.4 Environmental service conditions. The equipment shall maintain specified performance under the environmental conditions specified in 3.4.1 through 3.4.8.

3.4.1 Temperature. The BC and Cable Kit shall maintain specified performance in the operating temperature range of 0°Celsius (C) to 50°C. The equipment shall not be damaged nor shall operational performance be degraded when restored to the operating temperature range after being subjected to the nonoperating temperature range of -62°C to +71°C.

3.4.2 Humidity. The BC and Cable Kit shall maintain the specified performance when exposed to a relative humidity of 95 percent for both continuous and intermittent periods, including conditions wherein condensation occurs in and on the equipment in the form of both water and frost.

3.4.3 Salt fog (spray). The BC shall withstand the salt fog requirements of MIL-STD-810, Method 509.2.

3.4.4 Vibration. The BC shall withstand the vibration test specified in MIL-STD-167-1, Type I at frequencies up to 33 Hz.

3.4.5 Mechanical shock. The BC shall withstand the shock tests of MIL-S-901, Grade A, Class I, Type A.

3.4.6 Enclosure. The BC shall conform to the dripproof requirements of MIL-STD-108 when on an inclination of 45 degrees.

3.4.7 Inclination. The BC shall withstand the inclination test of 4.5.10.

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3.4.8 Electromagnetic compatibility (EMC). The BC shall operate compatibly in a submarine radio frequency environment. The BC shall conform to the requirements of MIL-STD-461, Part 6 for Class A5 equipment, as specified herein. Tests CE01 narrowband, CE03 narrowband, CS01, CS06 (at peak pulse of 400 V), RE01, RE02, RS01, RS02, and RS03 are applicable.

3.5 Parts, materials, and processes. Parts, materials, and processes shall be in accordance with the Parts, materials and processes paragraph of MIL-E-16400G, except as specified in 3.5.1 through 3.5.5. Nonstandard parts, materials, and processes shall require approval before use. Proprietary parts, materials, and processes shall be prohibited.

3.5.1 Parts. Selection and control of all parts shall be in accordance with MIL-STD-965, Procedure I. Integrated circuits (IC) conforming to MIL-M-38510, Class B, semiconductors conforming to MIL-S-19500, JANTX, and established reliability (quality level P) resistors and capacitors shall be used. Capacitors, semiconductor devices, resistors, and microelectronic devices shall be selected in accordance with MIL-STD-454, Requirement 2, Requirement 30, Requirement 33, and Requirement 64, respectively. Derating of electronic parts shall be accomplished in accordance with the requirements of AS-4613, Class B for a high ambient temperature of 50°C.

3.5.2 Unacceptable materials. Materials specified as unacceptable in MIL-E-16400G and materials specified in a through j shall not be used in the construction of the BC and Cable Kit:

a. Polychlorinated biphenyls
b. Mercury and mercury compounds (not to be used in any component, process, test, or inspection, or allowed to come into contact with the equipment at any time)
c. Asbestos and asbestos compounds
d. Fragile or brittle materials
e. Flammable or combustible materials
f. Beryllium and beryllium compounds
g. Lithium and lithium compounds
h. Cast iron, or 303 SE corrosion-resistant steel
i. Cadmium
j. Materials containing carcinogens

3.5.3 Interchangeability. The BC and Cable Kit shall conform to MIL-STD-454, Requirement 7.

3.5.4 Protective finish. All components of the BC shall be covered with a protective finish in accordance with MIL-F-18264 to provide corrosion resistance in a salt atmosphere. Aluminum shall not be left as a bare metal. The exterior and interior surfaces of the BC shall be painted in accordance with the requirements of the Protected equipment (interior use) paragraph of MIL-E-16400G.

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3.5.5 Printed circuit boards (PCBs). PCBs shall be fabricated in accordance with MIL-STD-454, Requirement 17, and shall conform to the requirements of MIL-STD-275 and MIL-P-55110. The copper clad laminate from which PCBs are made shall be in accordance with MIL-P-13949. This requirement applies to both single layer and multilayer boards. Connectors used with PCBs shall be fabricated in accordance with MIL-C-21097. Removal and insertion of printed circuit assemblies shall not damage the assemblies or any other part of the equipment. Each assembly shall be keyed to prevent incorrect insertion.

3.5.6 Safety criteria. BC design shall conform to the Safety criteria paragraph of MIL-E-16400G.

3.6 Electrostatic discharge protection (ESD) requirements. When metal oxide semiconductor parts and other parts sensitive to ESD, as specified in the Design protection paragraph of DoD-STD-1686 and DoD-HDBK-263 are used, the contractor shall incorporate protective circuitry in the BC design, ensure that ESD sensitive parts, modules, PCBs, subassemblies, and so forth, are protected in all phases of handling and testing in accordance with DoD-STD-1686, affix warning labels to protective packaging and to the equipment, provide warnings in all relevant areas of the equipment technical manual, and use identification marking of all visible ESD sensitive subassemblies (that is, on card edge) before handling of the equipment by maintenance personnel. The assembled unit shall be protected to a level of Class 2, as specified in DoD-HDBK-263.

3.7 Reliability. The mean-time-between-failures of the BC (θ_0 as defined by MIL-STD-781) shall be not less than 15,000 hours.

3.8 Maintainability. Maintainability shall be as specified in 3.8.1 through 3.8.3.

3.8.1 Corrective maintenance requirements, organizational level. The BC shall have a mean-corrective-maintenance-time (M_{ct}) not exceeding 0.5 hour and a maximum-corrective-maintenance-time (M_{maxct}) (95th percentile) not exceeding 1.5 hours when corrective maintenance is accomplished at the organizational level by the replacement of lowest assemblies (chassis-mounted parts and the power supply board). Corrective maintenance includes interchange, reassembly, alignment, and check out for all corrective actions. The BC is considered to be in a corrective maintenance mode whenever a failure (see 6.4) or malfunction occurs, including malfunctions that allow degraded operation.

3.8.2 Corrective maintenance requirements, depot level. The equipment shall be capable (inclusion of test points, and so forth) of repair within 24 hours at a depot facility. Corrective maintenance time includes preparation time; fault isolation; remove, repair, and replace time, or remove and replace time; warm-up time; and repair validation time.

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3.8.3 Preventive maintenance. With the exception of cleaning, the equipment shall not require preventive maintenance; cleaning shall not exceed 3 hours per calendar year.

3.9 Identification and marking. Identification and marking shall be in accordance with the identification and marking paragraph of MIL-E-16400G.

3.10 Workmanship. Workmanship shall be in accordance with 3.10.1 and 3.10.2.

3.10.1 General workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

3.10.2 Workmanship screen. The equipment shall withstand a defect detection vibration screen and temperature cycling in accordance with MIL-STD-2164 and as specified in 4.9.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to

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promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
   1. Production inspection (Group A) (see 4.4.1)
   2. Production control inspection (Group B) (see 4.4.2)
c. Inspection of packaging (see 4.10)

4.3 First article inspection. Unless otherwise specified (see 6.2), one unit shall be required for first article inspection. First article inspection shall consist of all examinations and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the examinations and tests specified in TABLE III.

TABLE III. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection	
				Group A	Group B
Performance	3.2	4.5.1	X	X	
Leakage current	3.2.5	4.5.2	X		X
Dimensions and weight	3.3.1	4.5.3	X	X	
Front panel	3.3.2	4.5.3	X	X	
Mounting plate heat sink	3.3.3	4.5.4	X		X
Human engineering	3.3.4	4.5.3	X	X	
Accessibility	3.3.5	4.5.3	X	X	
Cables and connectors	3.3.6	4.5.3	X	X	
Power interlock	3.3.7	4.5.3	X	X	
Temperature	3.4.1	4.5.4	X		X
Humidity	3.4.2	4.5.5	X		
Salt fog (spray)	3.4.3	4.5.6	X		
Vibration	3.4.4	4.5.7	X		X
Mechanical shock	3.4.5	4.5.8	X		
Enclosure	3.4.6	4.5.9	X		X
Inclination	3.4.7	4.5.10	X		

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TABLE III. Examinations and tests - Continued.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection	
				Group A	Group B
EMC	3.4.8	4.5.11	X		
Parts, materials, and processes	3.5	4.5.3, 4.5.12	X	X	
Safety	3.5.6	4.5.13	X	X	
ESD	3.6	4.6	X	X	
Reliability	3.7	4.7	X		X
Maintainability	3.8	4.8	X		
Identification and marking	3.9	4.5.3	X	X	
General workmanship	3.10.1	4.9.1	X	X	
Workmanship screen	3.10.2	4.9.2	X	X	

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.3.

4.4.1 Production inspection (Group A). Production inspection shall be conducted on every item offered for delivery. Production inspection shall comprise such examinations and tests which will prove the workmanship and reveal omissions and errors of the production process. Production inspection shall include functional and performance tests at a limited number of points, tests which detect deviations from design, tests of controls and adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests specified in Group A of TABLE III.

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in 4.4.2.1. Production control inspection shall consist of the examinations and tests throughout the entire range of operation; tests which detect deviations from design, tests of controls and adjustments, and tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of equipment to withstand this heat; tests of efficiency; and tests of the performance with other equipment in a system. The tests shall be performed on the complete BC as offered for delivery. Production control inspection shall include the examinations and tests specified in Group B of TABLE III.

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4.4.2.1 Sampling plan for Group B inspection. The BCs, submitted for Group B sampling tests, shall be selected by the contractor under the supervision of the Government quality assurance representative (QAR) and shall be representative of current production. Group B inspection, including sampling, shall conform to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. Group B inspection shall be performed on equipment that has passed Group A inspection specified in 4.4.1. The inspection levels shall be S-3 for normal, tightened, and reduced inspection. One failure shall cause rejection.

4.4.2.2 Rejected lot. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.2.3 Nonconforming production control sample BC. If a sample BC fails the inspection specified in 4.4.2, the contractor shall immediately investigate the cause of failure and shall report to the QAR the results thereof and details of the action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer or procuring activity.

4.4.3 Reinspection of conforming production control sample BC. Unless otherwise specified (see 6.2), sample BCs which have been subjected to, and have passed, production control inspection, may be accepted on the contract, provided they are resubjected to, and pass, production inspection after repair of all damage.

4.5 Test methods. The examinations and tests specified in TABLE III shall be conducted in accordance with the applicable test methods and quality assurance provisions of MIL-E-16400G, except as specified in 4.5.1 through 4.5.13. Testing shall be performed at ambient pressure and room temperature, unless otherwise specified herein.

4.5.1 Performance test. The BC and Cable Kit shall be subjected to a performance test including all operating controls to determine compliance with the performance requirements specified herein.

4.5.2 Leakage current test. The leakage current test shall be performed as specified in 4.5.2.1 through 4.5.2.3.

4.5.2.1 Primary power source. Leakage current shall be measured at the steady-state power line voltage and frequency of 120 V, 60 Hz.

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WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE BC DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.2.2 BC test condition. After power removal, each BC directly connected to an external power source and units deriving power from the BC shall be placed on an insulated surface. All safety ground conductors between the BC and units deriving power from the BC shall be intact. The safety ground conductor between the equipment and the source power shall be disconnected during the test. OBSERVE WARNING STATEMENT. The BC shall be connected as shown in FIGURE 5 for single-phase source power.

4.5.2.3 Measurement. Leakage current shall be measured on the BC in its normal operating configuration. BC controls in each operating mode shall be such that maximum power will be utilized during leakage current measurement. The leakage current shall be determined by the voltage-drop method. A true root-mean-square (rms) voltmeter shall be used. The voltage measured across the 1500-ohm resistor, when equal to 7.5 V, represents 5 mA of leakage current. The overall measurement error shall not exceed 5 percent. The probe shall be used on all external conducting parts. The voltage shall be measured from each part to ground for every combination of switch positions available in FIGURE 5. The disconnected safety ground conductor shall be reconnected immediately after the test is completed.

4.5.3 Surface examination. The BC and Cable Kit shall be examined to ensure compliance with the applicable referenced specification paragraphs with respect to:

a. Dimensions and weight (see 3.3.1)
b. Front panel (see 3.3.2)
c. Human engineering (see 3.3.4)
d. Accessibility (see 3.3.5)
e. Cables and connectors (see 3.3.6)
f. Power interlock (see 3.3.7)
g. Parts, materials, and processes (see 3.5)
h. Identification and marking (see 3.9)

In the process of examination, the item shall not be disassembled in a manner that will affect the performance, durability, or appearance of the item. The examination shall include a check of all operating controls, circuit functions, test provisions, and adjustments.

4.5.4 Temperature tests. The BC shall be subjected to low and high temperature tests in accordance with MIL-STD-810, Method 501.2 and Method 502.2, to determine conformance to the requirements specified in 3.4.1.

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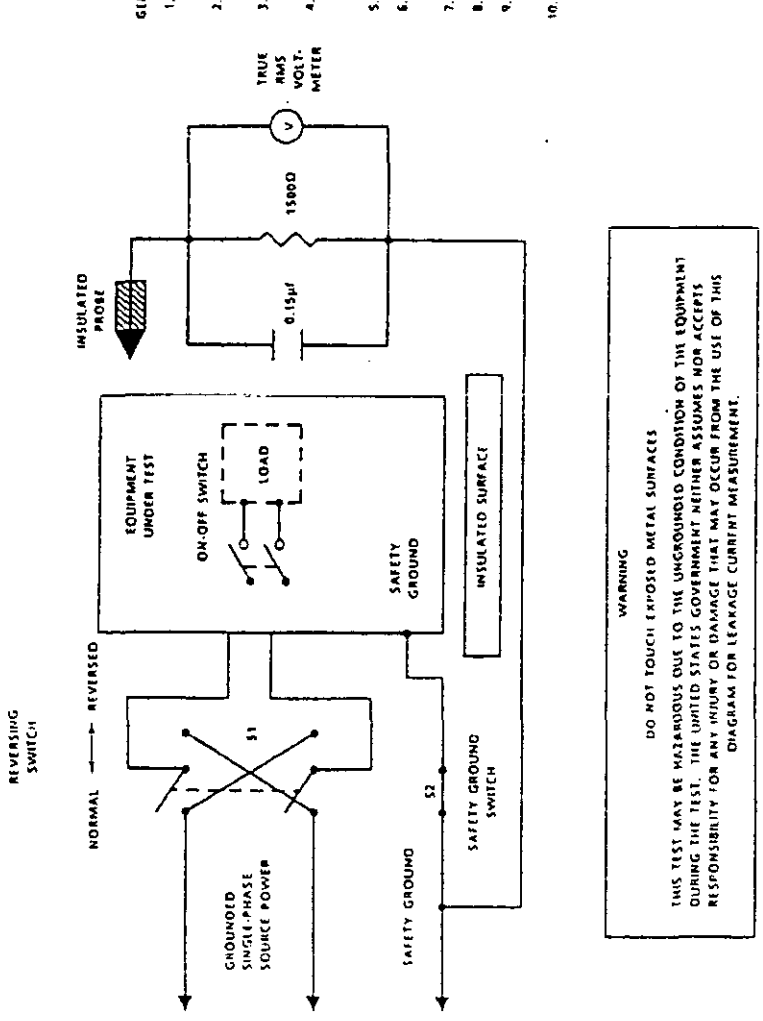


FIGURE 5. Single-phase test diagram for leakage current measurement.

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4.5.5 Humidity test. The BC shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507.2 to determine conformance to the requirements specified in 3.4.2.

4.5.6 Salt fog. The salt fog test specified in MIL-STD-810, Method 509.2 shall be applied to the finishes and coatings on parts, and frame and enclosure structures as finally assembled for use. The test shall not be applied to the complete equipment.

4.5.7 Vibration test. The BC shall be hard-mounted to the vibration table using a test fixture and shall be subjected to the amplitude inputs of MIL-STD-167-1, Vibratory displacement of environmental vibration table, for each of the three orthogonal axes (principal axes of BC). The frequency range shall be between 4 Hz and 33 Hz. The BC shall bolt to the test fixture identical to the bulkhead mounting design. The resonance frequency of the fixture shall be greater than 75 Hz. The BC shall conform to the performance requirements specified herein, before, during, and after the test.

4.5.8 Mechanical shock. The BC shall be hard-mounted to the shock table using a test fixture and shall be subjected to the shock inputs of MIL-S-901 for Grade A, Class I, Type A equipment. The BC shall be mounted to the test fixture identical to the bulkhead mounting design. The low weight, high impact shock table shall be used for the test. The BC shall be energized and operating conditions checked after each shock blow. The BC shall conform to the performance requirements specified herein, after the test.

4.5.9 Enclosure. The BC shall be tested in accordance with MIL-STD-108 to determine conformance to the requirements specified in 3.4.6.

4.5.10 Inclination test. The BC shall be energized and fully operating during the test. The BC shall be inclined at the rate of five cycles to seven cycles per minute in one plane to angles of 60 degrees on either side of vertical for a period sufficiently long to determine the characteristics under such motion or for a minimum of 30 minutes. The test shall be repeated with the BC reoriented 90 degrees to the plane in which it was originally tested. At the conclusion of these cyclic tests, the cyclic motion shall be stopped and the inclination adjusted to an angle of 15 degrees. The BC shall be then operated for a sufficient period to ensure that continuous operation can be maintained. The BC shall be then rotated through vertical to 15 degrees in the opposite direction, and the test for continuous operation shall be repeated.

4.5.11 EMC. Conformance to the requirements specified in 3.4.8 shall be verified by tests performed in accordance with MIL-STD-462. Interface connections with actual or simulated loads shall be utilized during testing.

4.5.12 Parts. Parts rescreening shall be implemented on all Military semiconductor devices.

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4.5.12.1 ICs. ICs shall be inspected as specified in a through c:

a. 100 percent inspection
b. Measure input/output logic
c. Measure over temperature range 120°C (or the maximum component operating temperature, whichever is less) to 25°C

4.5.12.2 Transistors. Transistors shall be inspected as specified in a and b:

a. Measure breakdown voltages (collector to emitter, collector to base, and base to emitter)
b. Measure Hfe (minimum and maximum) (small signal short circuit forward current transfer ratio, common emitter (minimum and maximum)); Ieo (emitter cutoff current, direct current (DC), collector open); Icbo (collector cutoff current, DC, emitter open); and Iceo (collector cutoff current, DC, base open)

4.5.13 Safety verification. Examinations, analyses, and tests shall be performed to verify conformance to 3.5.6.

4.6 ESD testing. All ESD-susceptible equipment shall be tested for compliance with the requirements specified in DoD-STD-1686. Testing done prior to first article testing for component classification shall be considered destructive and parts tested shall not be used in equipment delivered to the Government.

4.7 Reliability verification test. The BC shall withstand a reliability test conducted in accordance with MIL-STD-781 and the test conditions specified in a through f:

a.	Test time:	1500 hours
b.	Accept criterion:	No failure
c.	Reject criterion:	One or more failures
d.	Electrical stress:	Prime power voltage cycling shall be 0.33 at nominal minus 7 percent, 0.33 at nominal, and 0.33 at nominal plus 7 percent, repetitive every hour.
e.	Thermal stress:	Three cycles per day nominal
f.	Moisture (humidity):	Ambient

4.8 Maintainability equipment demonstration. Conformance to the requirements for maintainability M_{ct} and M_{maxct} (95th percentile) shall be determined by an organizational level maintainability demonstration in accordance with MIL-STD-471, Method 9. A total of 25 candidate corrective maintenance faults shall be determined in accordance with APPENDIX A of MIL-STD-471. The demonstration shall consist of a total of five corrective maintenance tasks, each requiring replacement of a failed subassembly or chassis-mounted part. The procuring activity or authorized representative

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shall determine a sample for the demonstration from the candidate faults. In addition, all preventive maintenance tasks shall be performed in accordance with the applicable procedures.

4.9 Workmanship. Workmanship shall be as specified in 4.9.1 through 4.9.2.2.

4.9.1 General workmanship. The BC and Cable Kit, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process and at the end item level for conformance to the requirements of 3.10.1. Each solder connection and its associated wiring or leads shall be visually examined.

4.9.2 Workmanship screen. Vibration and temperature cycling shall be performed on each BC. Vibration shall be performed prior to temperature cycling. The vibration may be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration.

4.9.2.1 Vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 6. Input vibration levels shall be measured at the mounting points of the item under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz, and 100 Hz, for vibration frequencies above 500 Hz, shall be used for the control and analysis of the acceleration spectral density (ASD). The instantaneous acceleration peaks shall be limited to three times the rms acceleration level. The item shall be energized with appropriate input signals applied during the vibration to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed.

4.9.2.2 Temperature cycling. Each BC shall be subjected to 10 cycles of the temperature curve shown in FIGURE 7. The temperature rate of change shall be not less than 5°C per minute. BC power shall be turned on and off at the indicated times. The BC shall be positioned for maximum exposure to the changing temperature. Where performance measurements are specified, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the BC reaches 80 percent of the chamber temperature. When failure occurs, the BC shall be repaired and the cycling continued for a cumulative total of 10 cycles.

4.10 Inspection of packaging. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

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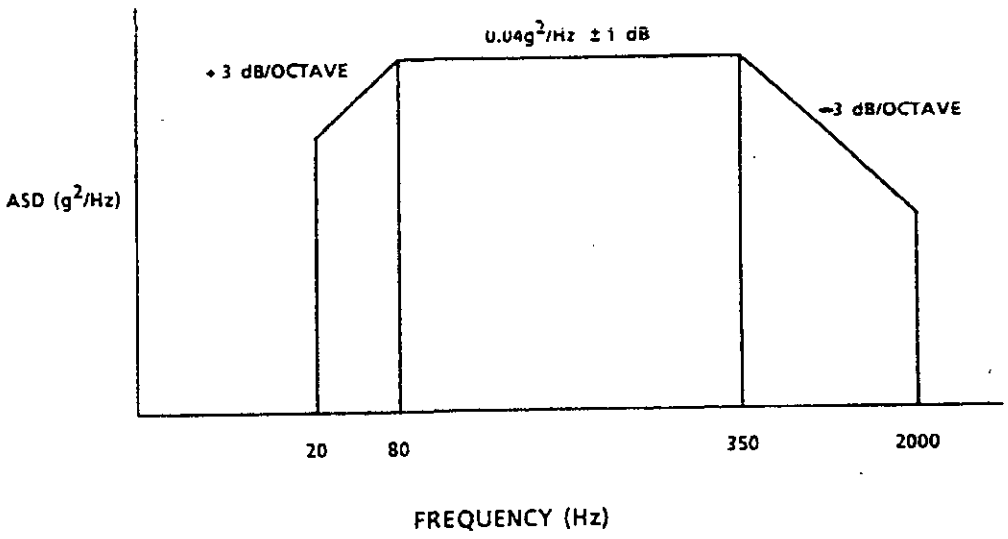


FIGURE 6. Random vibration curve.

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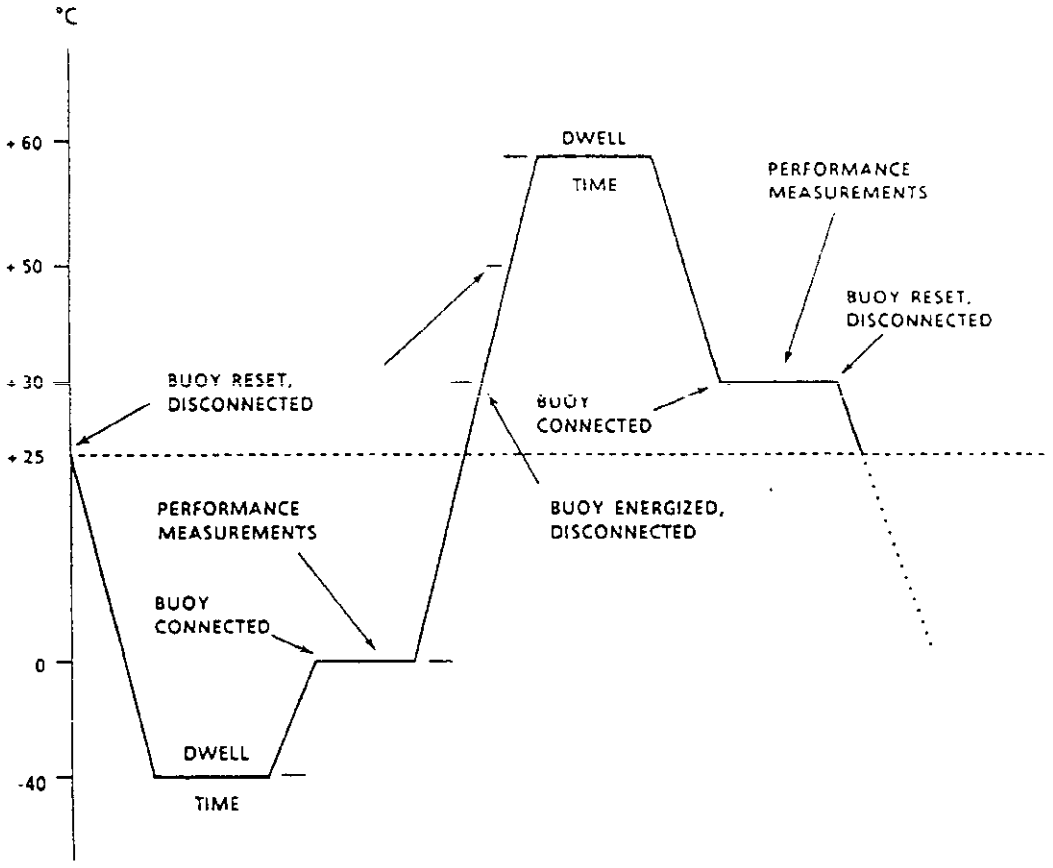


FIGURE 7. Random temperature curve.

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5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The BC and Cable Kit, covered by this specification, are intended for use on board all SSN and SSBN submarines as a part of the Buoy Transmitting Set, AN/BRT-6.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. When reinspected production control sample units may not be accepted (see 4.4.3)
d. Levels of preservation, packaging, packing, and marking required (see 5.1)

6.3 First article. When a first article inspection is required, the item should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangement for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definition of failure. A failure is the inability to conform to the performance requirements specified herein.

6.5 Subject term (key word) listing.

Charger, Battery, PP-7806/BRT-6
Intended use: On board SSN and SSBN submarines
PP-7806/BRT-6, Charger, Battery

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6.6 Supersession data. This specification includes the requirements of ELEX-B-596, dated 3 February 1984.

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NAVY - EC

(Project No. 5820-N857(EC))

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a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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